Early Prediction of Student Outcomes Using Structured Institutional Data

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***Abstract*—Student attrition remains a persistent challenge in higher education, with consequences for learners (lost time and debt burden) and institutions (reduced completion rates and inefficient resource planning). This paper studies early prediction of three student outcomes—dropout, continued enrollment, and graduation—using structured institutional data that includes academic progress indicators (e.g., first-semester performance), financial status (e.g., tuition payment and debt), and demographic/background attributes.**

We frame the task as a supervised multiclass classification problem and evaluate a baseline pipeline of data preprocessing, feature selection, and comparative modeling using standard machine learning classifiers. Our current results are preliminary: using the course dataset described in our project, we obtain up to 77.9% accuracy on held-out data.

The main contributions are: (i) a problem formulation and feature taxonomy aligned with advising decisions, (ii) an empirical comparison of classical machine learning baselines for early-risk screening, and (iii) a discussion of practical gaps—generalization across cohorts and interpretability—that motivate the next stage of this work.

***Index Terms*—student attrition, dropout prediction, early warning systems, multiclass classification, educational data mining, academic success, institutional data.**

I. Introduction

Student dropout is a major operational and societal problem for colleges and universities. Institutions seek actionable, early indicators of risk so that advisors can intervene before academic or financial difficulties become irreversible. Prior work shows that administrative records and early academic signals can support accurate prediction of attrition in higher education [1, 2].

A. Problem statement

Given student records containing academic, financial, and background variables observed at enrollment and during the first academic term(s), the goal is to predict a student's eventual outcome: dropout, continued enrollment, or graduation. We target early prediction to support timely intervention.

B. Research challenges

Key challenges include: (i) feature availability and timing (many strong predictors appear only after substantial progression), (ii) class imbalance and heterogeneous student populations, and (iii) model generalization across cohorts and programs, where performance can degrade due to dataset shift [3].

C. Research objectives and approach

This work aims to (1) identify which academic and financial features are most predictive of risk, and (2) establish transparent baseline models suitable for an early-warning workflow. We implement a standard supervised learning pipeline—data cleaning, train/validation/test splitting, model training, and evaluation—and compare multiple classifiers (e.g., logistic regression and tree-based ensembles), following common practice in student success modeling [4].

D. Contributions

The contributions of this paper are:

- A clear early-warning formulation for multiclass outcome prediction (dropout/enrolled/graduated).
- A feature-driven analysis of academic progress versus financial and background variables.
- A baseline experimental benchmark (preliminary) to guide subsequent improvements and interpretability work.

E. Paper organization

Section II reviews related work on dropout prediction and early-warning systems. Subsequent sections (to be completed) will describe the dataset, methods, experiments, and discussion.

II. Related Work

Research on student outcome prediction spans traditional higher education settings and large-scale online learning environments.

A. Dropout prediction in higher education

Aulck *et al.* model college attrition using large-scale administrative records and show that meaningful prediction is possible even from early transcript information [1]. Lakkaraju *et al.* propose a machine learning framework aimed at identifying at-risk students sufficiently early to enable intervention, emphasizing operational constraints and the need for actionable models [2]. More recent work expands the scope to multiclass success outcomes and public benchmarks; for example, the "Predicting Student Dropout and Academic Success" dataset and accompanying study provide a widely used reference task for comparing classifiers under a consistent feature set [4].

B. MOOC dropout prediction and evaluation pitfalls

In MOOCs, dropout rates are high and clickstream data enable fine-grained behavioral features. Kloft *et al.* use weekly activity features and support vector machines to predict dropout between course weeks [5]. Evaluation methodology can lead to overly optimistic estimates when models are trained and tested on the same course; Whitehill *et al.* highlight this issue and argue for more realistic cross-course evaluation setups [3].

C. Gaps and positioning of this work

Across both higher education and MOOCs, two gaps recur: (i) early prediction with features available before performance collapse, and (ii) deployment-oriented considerations such as interpretability and stability under cohort shift. Our work positions itself as an early-warning baseline study on structured institutional data, emphasizing feature timing (academic progress vs. financial status) and establishing a reproducible benchmark that can be extended with interpretability analyses and robustness checks.

III. Proposed Algorithm

This section describes the end-to-end baseline pipeline used to predict student outcomes (Dropout/Enrolled/Graduate) from structured institutional variables, aligned with the accompanying project portfolio [6].

A. Data and Feature Groups

We use the public "Predicting Student Dropout and Academic Success" dataset [4]. Variables are grouped into: (i) academic progress (e.g., first-semester grades and approved/passed units), (ii) financial status (e.g., tuition fees up to date, debt), (iii) demographics/background (e.g., age at enrollment, marital status), and (iv) contextual indicators (e.g., unemployment rate).

B. Preprocessing and Data Quality Checks

Following the documented cleaning checklist in the project work [6], we (i) validate ranges (e.g., admission grade on a 0–200 scale; semester grades on a 0–20 scale; binary fields limited to 0/1), (ii) standardize categorical coding into readable labels (Dropout/Enrolled/Graduate), and (iii) handle missing or invalid values with documented rules (median imputation for numeric fields when appropriate; removal only when target is missing).

C. Model Families

We treat the task as supervised multiclass classification. To establish transparent baselines and quantify nonlinearity, we compare ten standard classifiers using the same split protocol [6]: (i) a majority-class baseline, (ii) logistic regression, (iii) support vector machine, (iv) naive Bayes, (v) k-nearest neighbors, (vi) decision tree, (vii) gradient boosting, (viii) extra trees, (ix) histogram gradient boosting, and (x) random forest.

D. Overall Procedure

The baseline workflow follows these steps [6]:

1) Profile and validate the dataset (ranges, types, missingness, category consistency).
2) Clean and transform fields; encode categorical variables; scale numeric variables as needed.
3) Split the data into train/test (holdout).
4) Train candidate models on the training split (with tuning when applicable).
5) Select the best-performing model by holdout accuracy and stability.
6) Report accuracy and a confusion matrix for error analysis.

IV. Experiments

This section documents the data, preprocessing, baselines, tuning, metrics, main findings, ablations, robustness, and reproducibility. The guiding principles are fair comparison and practical relevance.

A. Datasets and Prediction Task

Two public files are used: `googleplaystore.csv` (metadata and target `Rating`) and `googleplaystore_user_reviews.csv` (free-form reviews). Records are joined on `App` after normalizing whitespace and case (e.g., trimming, lowercasing, resolving obvious duplicates). The task is app-level regression: predict an app's numeric rating from metadata and aggregated sentiment.

B. Preprocessing Details

Cleaning & typing: `Installs`/`Reviews` remove symbols (+, commas), `Price` parses `$`-prefixed strings, `Size` converts `k`/`M` to MB. Out-of-range ratings are discarded. **Transforms:** heavy-tailed counts use $\log(1+x)$, then standardization with training-only statistics. **Categoricals:** one–hot encoding with an "other" bucket for rare/unknown values; multi-genre entries are split on the delimiter and multi-hot encoded. **Sentiment:** VADER per review, then per–app means by default; length-weighted means and dispersion are evaluated in ablations. Apps without reviews are either assigned neutral defaults or dropped (both variants reported).

C. Train/Validation/Test Protocol

A stratified split on binned ratings preserves distribution: 70% train, 15% validation, 15% test. Hyperparameter selection uses $k = 5$ folds within training data; early stopping monitors validation loss with a patience of 50 rounds. All preprocessing (standardization statistics and one–hot vocabularies) is fit on training folds only and applied to validation/test to prevent leakage.

V. Experiments

A. Dataset

Experiments use the "Predicting Student Dropout and Academic Success" dataset [4]. The project implementation summarizes 4,424 student records with 36 variables and a three-class target (Dropout/Enrolled/Graduate) [6].

TABLE I Best baseline performance on the held-out test split.

Model	**Accuracy (Test)**
Random Forest	0.779

B. Protocol and Metrics

We use a standard train/test holdout and report multiclass accuracy as the primary metric. To understand failure modes, we also inspect the confusion matrix for the best model [6].

C. Results

Across the compared baselines, the random forest achieves the best held-out accuracy of 77.9% [6]. The confusion-matrix analysis suggests most errors occur between Enrolled and Graduate, while Dropout is comparatively easier to identify given stronger academic and financial signals [6].

D. Limitations

Results are preliminary. In realistic deployments, model performance can degrade under cohort shift and institutional changes, motivating more robust evaluation and interpretability analyses [3].

VI. Conclusion

This paper studies early prediction of student outcomes (Dropout/Enrolled/Graduate) using structured institutional data. Using the dataset of [4] and the baseline pipeline summarized in the associated project portfolio [6], classical machine learning baselines achieve up to 77.9% held-out accuracy.

Next steps focus on (i) validating generalization across cohorts/programs, (ii) incorporating more explicit feature-timing constraints for "early" prediction, and (iii) adding model explanations that support advisor decision-making.

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